

Multimodal Representation Learning for Clinical Decision Support from Vision and Tabular Data

Maxime Di Folco

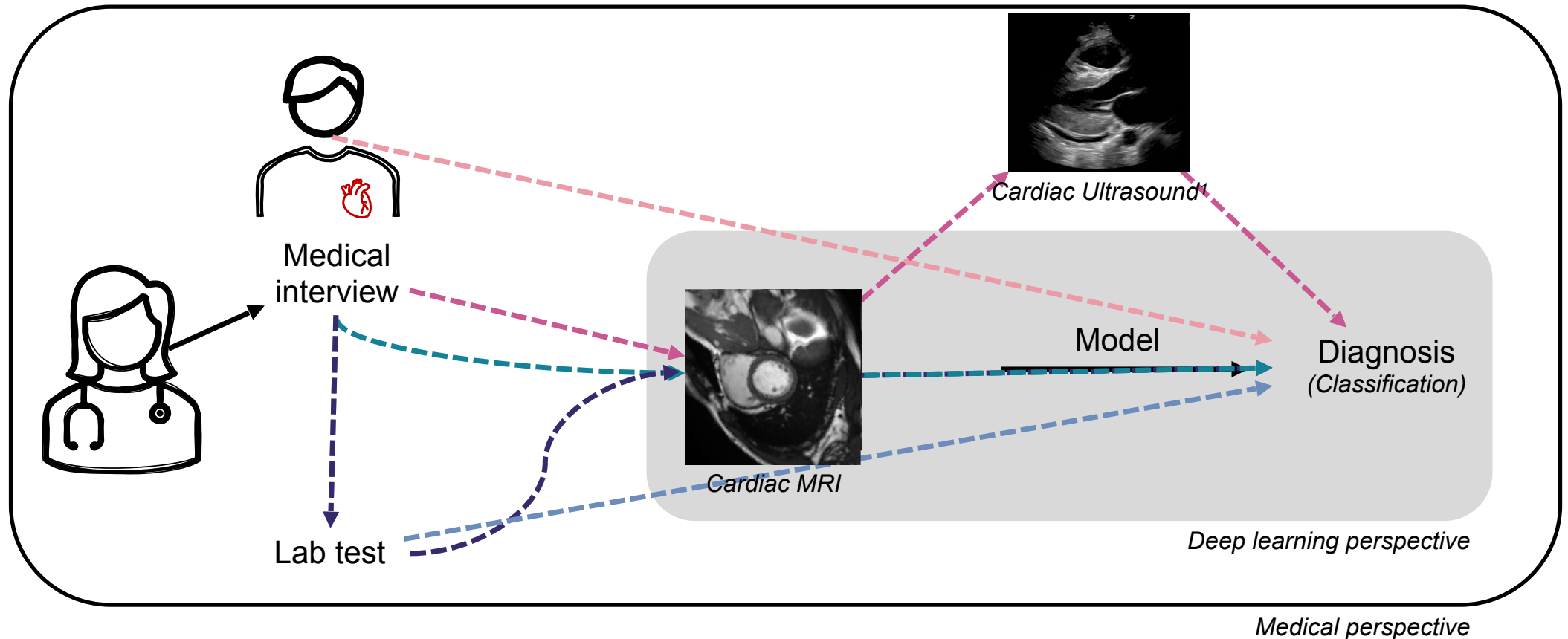
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JDSE 2026 – 24/09/2026



Context

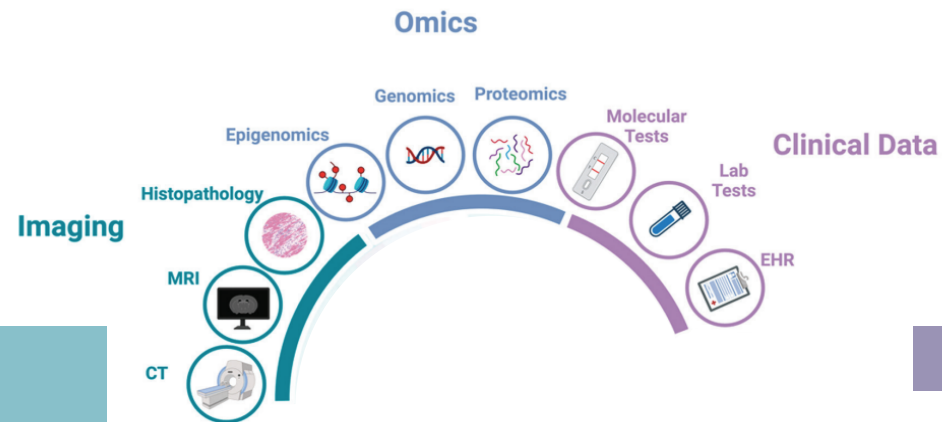
Images are not taken in isolation,
they are answers to clinical questions



1 – Image from <https://www.pocus101.com/cardiac-ultrasound-echocardiography-made-easy-step-by-step-guide/>

Context

Medicine is a multimodal problem



**Multimodality with
Heterogeneous data**

High-stakes decision

1 - Simon et al., DIR (2025)

Context

Deep learning relies on data

Biobanks are a great source of resources

UK Biobank [1]

- Around 500,000 participants
- 50 000 with imaging

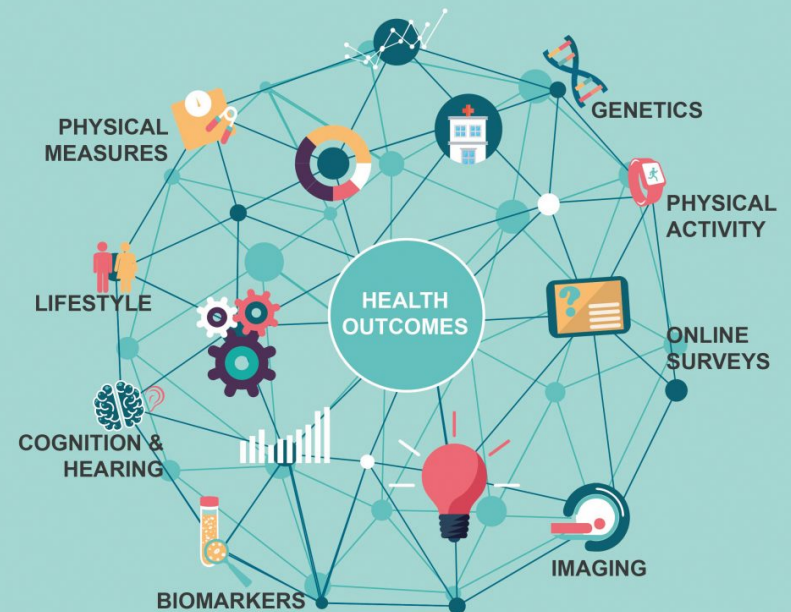


→ Stored in the tabular format

[1] Littlejohns, T. J., Holliday, J., Gibson, L. M., Garratt, S., Oesingmann, N., Alfaro-Almagro, F., ... & Allen, N. E. The UK Biobank imaging enhancement of 100,000 participants: rationale, data collection, management and future directions. *Nature communications*, 2020

BREADTH AND DEPTH

A summary of all the information gathered and available for research can be found in the UK Biobank Data Showcase.



<https://www.scientia.global/uk-biobank/>

Limitations of Biobank-Driven Learning

Biobanks:

Large-scale, well-curated multimodal data

- Designed for research, not clinical decision-making
- High completeness, low missingness
- Mostly healthy populations
- Distribution mismatch with clinics

Clinics:

Limited and fragmented data

- Design for patient care
- Frequent missing modalities
- Mostly diseased patients

We need methods that can **train using several modalities** and **leverage biobanks**, but **infer from only modalities available** in a clinical scenarios

Presentation overview

Combining Medical Imaging and Clinical Data

Tables Guide Vision

Hasny et al., WACV (2025)

**Robust Vision-Tabular
Framework**

Hasny et al., ICML (2026)

Towards real clinical application

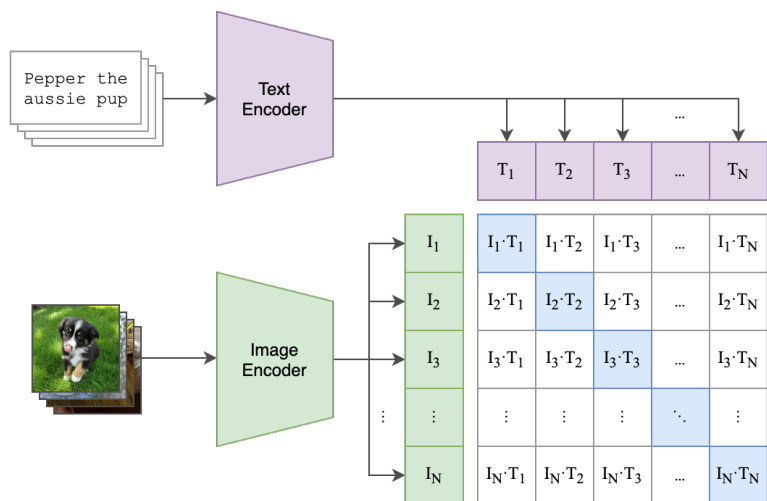
**Risk Estimation of
Cardiovascular Diseases**

Under review

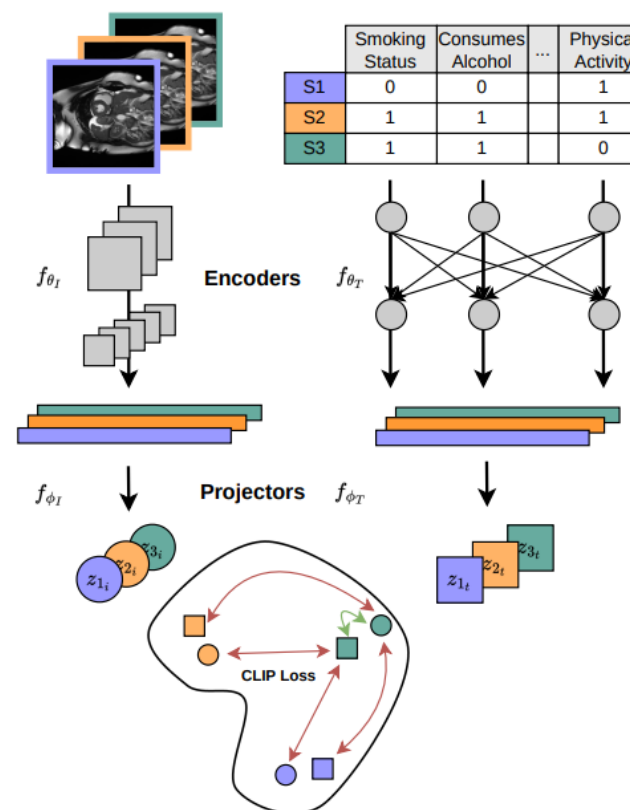
Typical Multimodal Learning Approaches

Language Supervision (CLIP [1])

(1) Contrastive pre-training



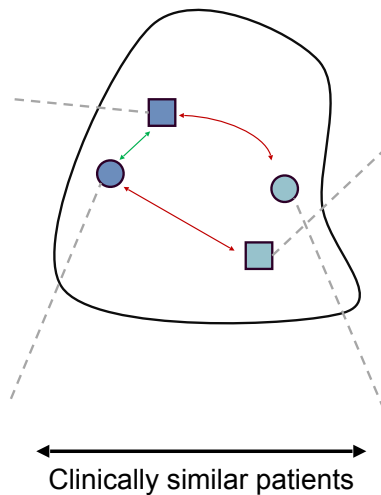
Tabular Supervision (MMCL [2])



[1] Radford, A., Kim, J. W., Hallacy, ... & Sutskever, I. Learning transferable visual models from natural language supervision. In *International conference on machine learning*. 2021

[2] Hager, Paul, Martin J. Menten, and Daniel Rueckert. "Best of both worlds: Multimodal contrastive learning with tabular and imaging data." *CVPR2023*.

Typical Contrastive Learning Approaches



False negative
table

Typical contrastive learning

one-to-one pairing



ignores clinical similarity

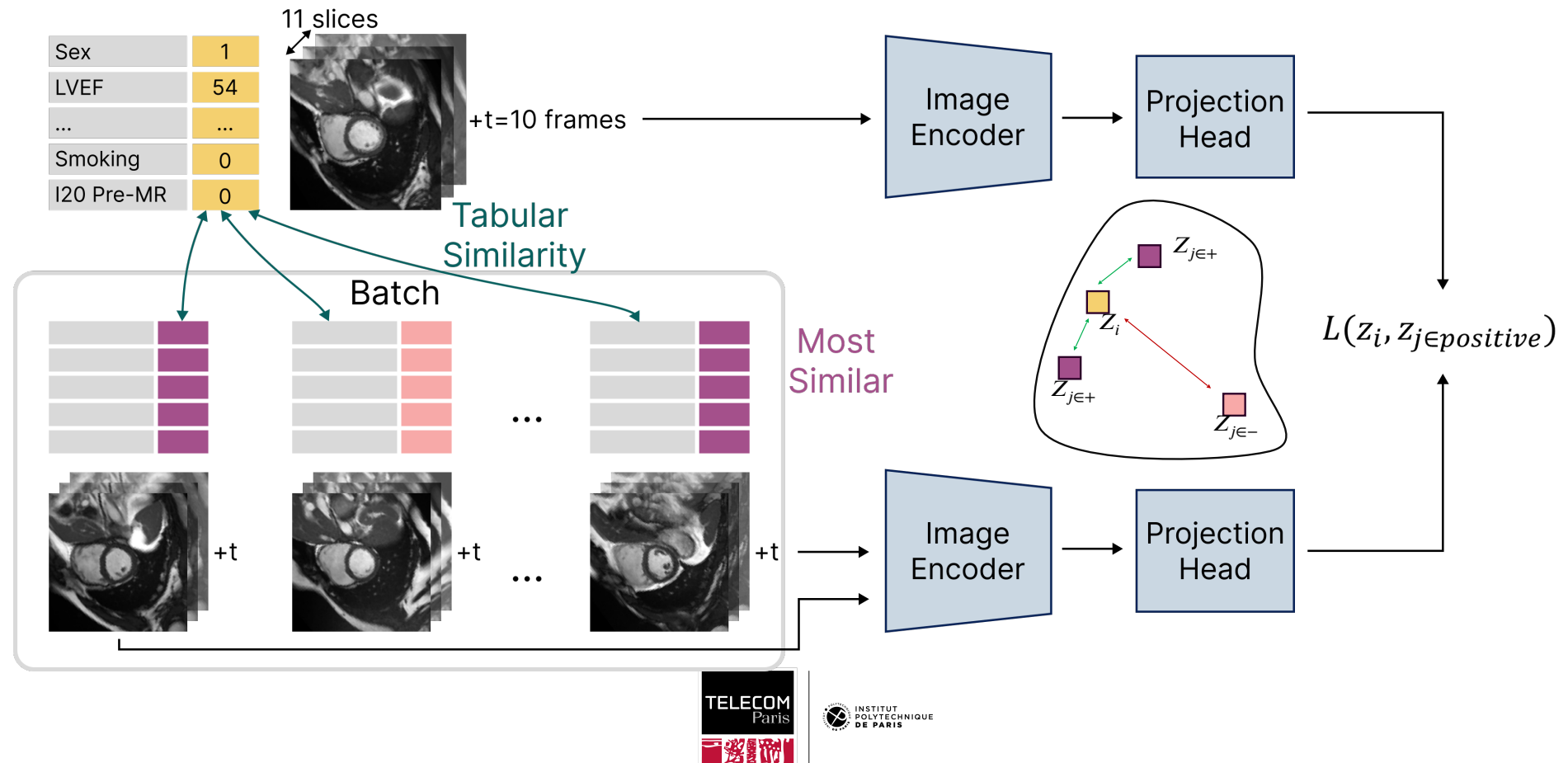


false negatives

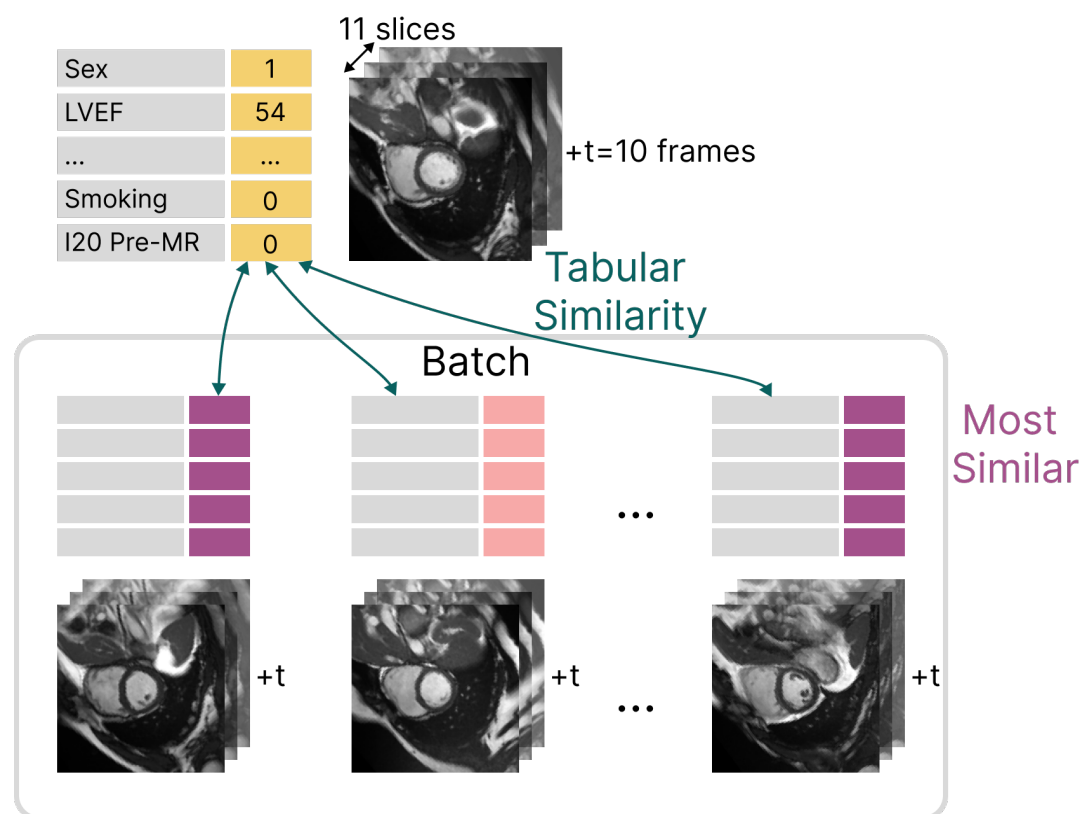
Combining Medical Imaging and Clinical Data

TGV: Tables Guide Vision

Hasny, M., Di Folco, M., Bressen, K., & Schnabel, J. Tables guide vision: Learning to see the heart through tabular data. *In Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision. 2025*



TGV: Tables Guide Vision



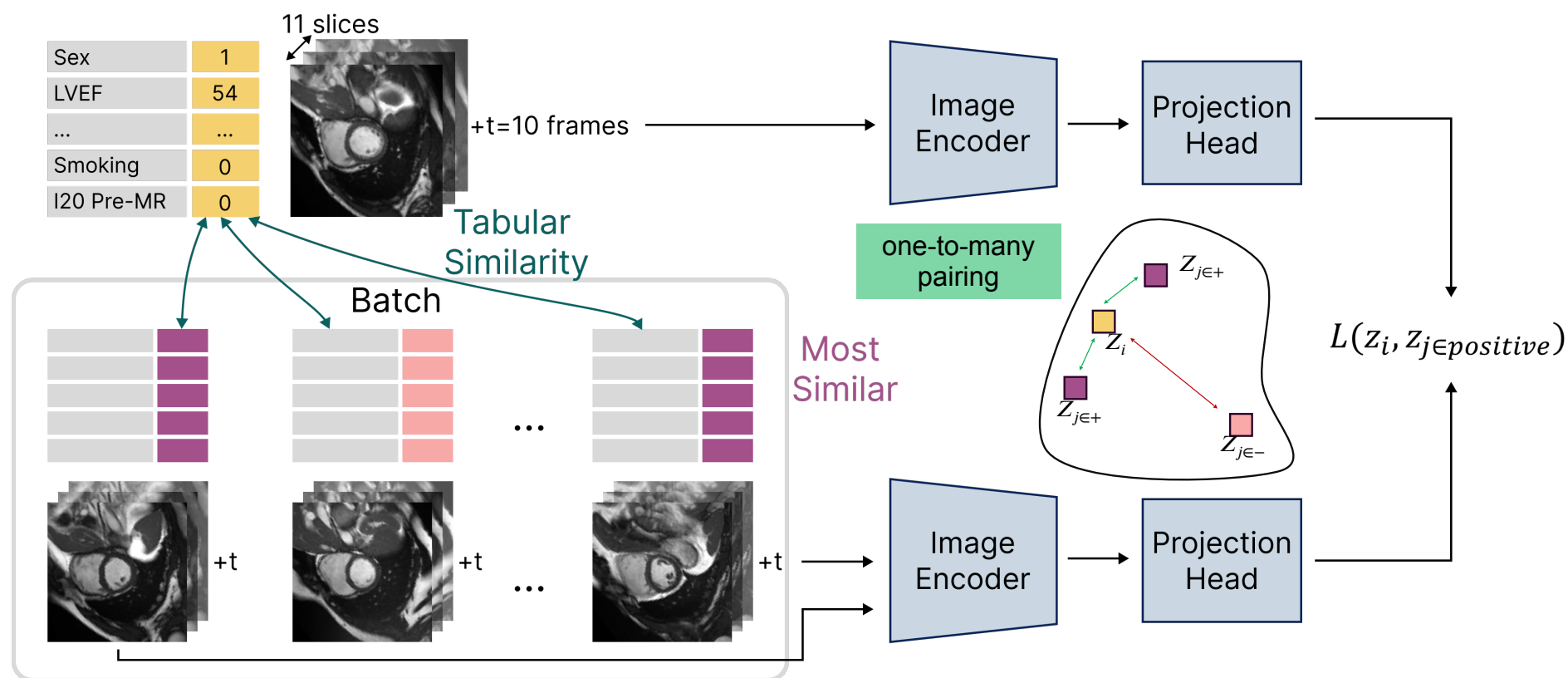
$$S = \lambda S_{con} + (1 - \lambda) S_{cat}$$

- Categorical similarity matrix S_{cat}
 - Cosine similarity
- Continuous similarity matrix S_{con}
 - Euclidean distance
- Weighting factor λ

Experiment settings:

- Cardiac MRI + 24 Cardiac attributes (10 categorical, 14 continuous)

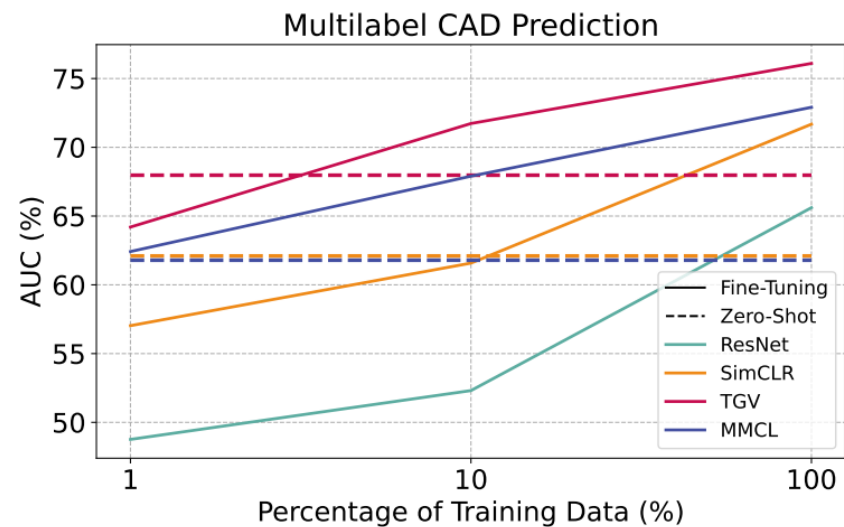
TGV: Tables Guide Vision



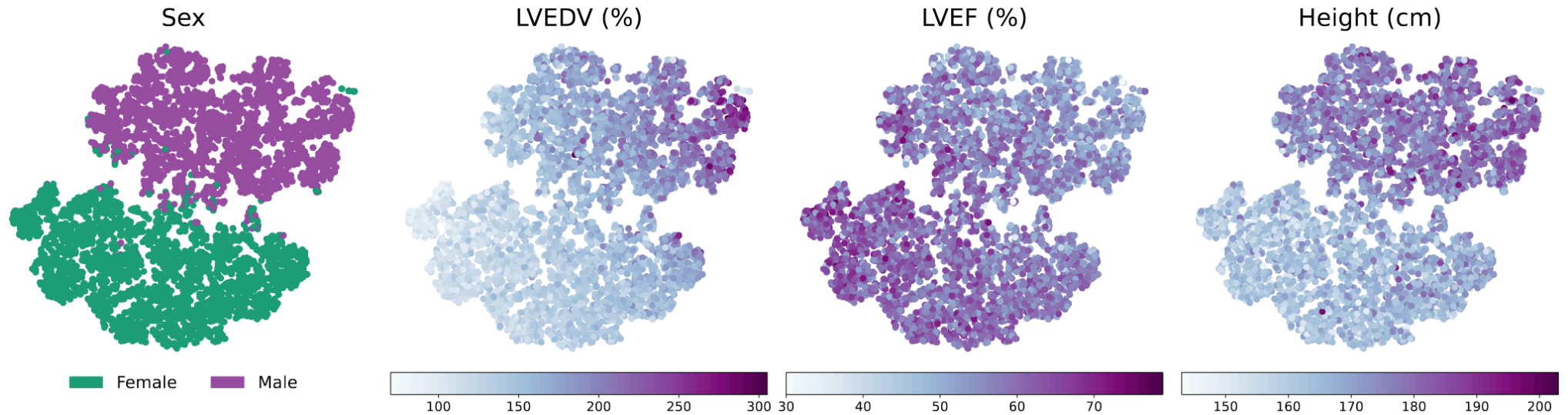
Classification of Cardiac Diseases

AUC		
Model	CAD $\uparrow$	
	ZS	FT
Mean-Guess	-	-
<i>Supervised</i>		
ResNet50 [19]	-	65.61
<i>Image Augmentation</i>		
SimCLR [6]	62.05	71.68
BYOL [16]	56.99	67.32
SimSiam [8]	57.01	69.89
Barlow Twins [40]	55.12	65.01
<i>Tabular Supervision</i>		
MMCL [18]	62,49	<u>72.91</u>
<i>Tabular Guidance</i>		
TGV (Ours)	68.70	76.1

Two scenarios: Zero-Shot (ZS), Fine-Tuning (FT)



Clinical relevant representation



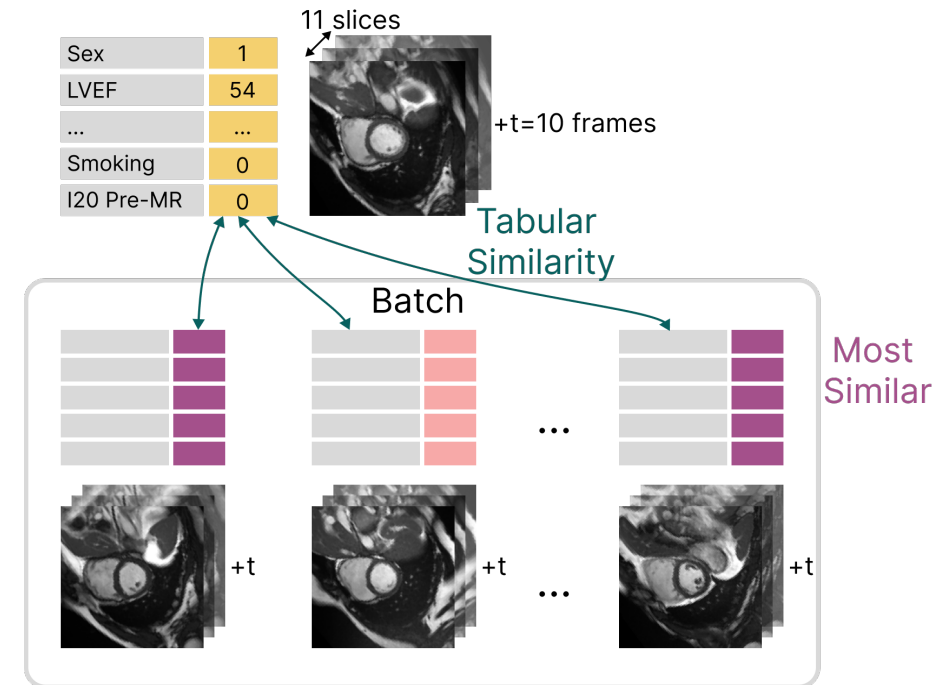
LVEDV: Left Ventricular End-Diastolic Volume; LVEF: Left Ventricular Ejection Fraction

Conclusion

- **Tabular data** to create **clinically meaningful training pairs** for visual representation learning
- **Tabular guidance** achieved superior results as **tabular supervision**

Limitations:

- Noisy guidance with more attributes (e.g. Lifestyle, Diet & Nutrition, ...)
- No multimodal inference



Presentation overview

Combining Medical Imaging and Clinical Data

Tables Guide Vision

Hasny et al., WACV (2025)

Robust Vision-Tabular
Framework

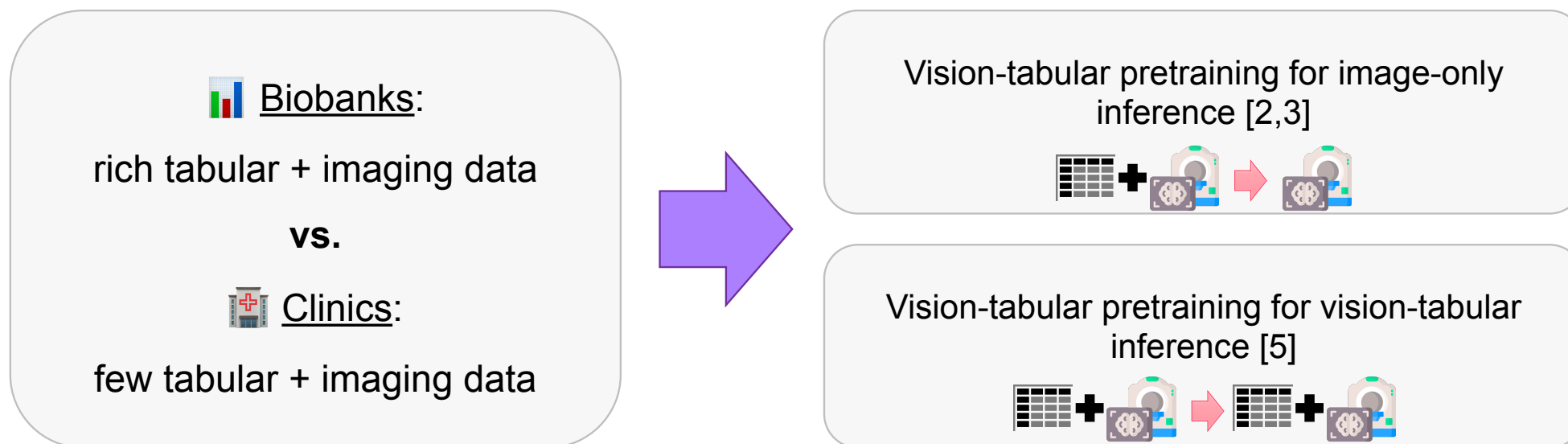
Hasny et al., ICML (2026)

Towards real clinical application

Risk Estimation of
Cardiovascular Diseases

Under review

Biobank-Driven Learning



[2] Hager, Paul, Martin J. Menten, and Daniel Rueckert. "Best of both worlds: Multimodal contrastive learning with tabular and imaging data." CVPR2023.

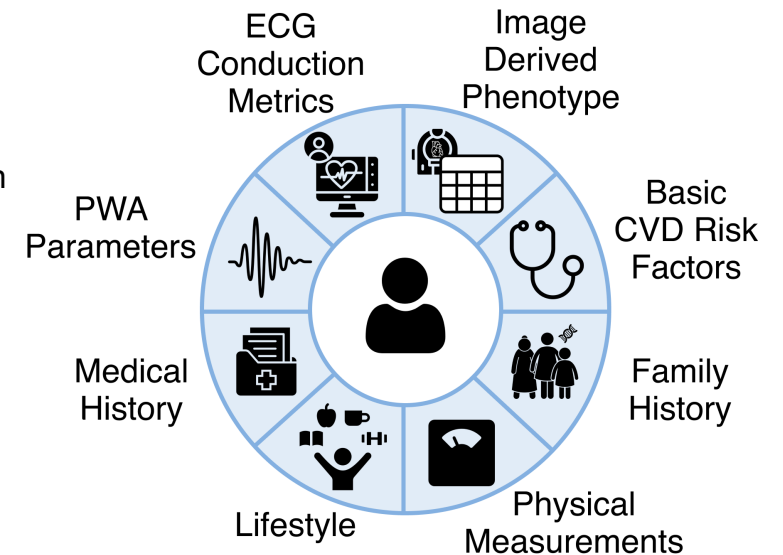
[3] Hasny, Marta, et al. "Tables guide vision: Learning to see the heart through tabular data." WACV2026.

[5] Du, Siyi, et al. "TIP: Tabular-image pre-training for multimodal classification with incomplete data." ECCV2024.

Adding more attributes

129 attributes describing:

- **Lifestyle:** Alcohol drinker status, current tobacco smoking, ...
- **Diet & Nutrition:** Body fat percentage, ...
- **Cardiac Indicators:** Systolic blood pressure, Left Ventricular Ejection Fraction
- **Physical Measurements:** Height, Waist circumference, ...
- **Health & Medical History:** Heart Failure, Diabetes, ...
- **Demographic Data:** Age, Sex



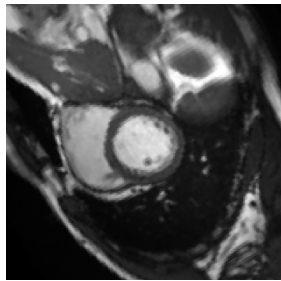
Combining Medical Imaging and Clinical Data

Limitations of Vision-Tabular Learning



Augmentation-induced misalignment

- But what happens when you corrupt tabular features in multimodal vision-tabular learning?



Ground Truth Pair

Tabular Data
Corruption

Non-factual Pair

Missingness
Simulation

Factual Pair

Combining Medical Imaging and Clinical Data

Monotonicity & Full-Spectrum Robustness Gaps

Combining Medical Imaging and Clinical Data

RoVTL: Robust Vision-Tabular Learning

Hasny, M., Daza, L., Bressem, K., Di Folco, M., & Schnabel, J. No Data? No Problem: Robust Vision-Tabular Learning with Missing Values. *International Conference on Machine Learning*. 2026



Tabular ranking loss



Tabular ranking loss

- Goal: encourages the network to perform better when more tabular data are available

$$L_{tab} = \max(L_{task}(i, t^+) - L_{task}(i, t^-), 0) \longrightarrow \text{adapted from [7] to the tabular format}$$

$$L_{multi} = L_{task}(i, t^+) + L_{task}(i, t^-) + \lambda L_{tab}$$

Why prioritize the case where more data is available?

→ Addressing the monotonicity gap

[7] Li, S., Chen, C., & Han, J. (2025). Simmlm: A simple framework for multi-modal learning with missing modality. *IEEE/CVF International Conference on Computer Vision. 2025*

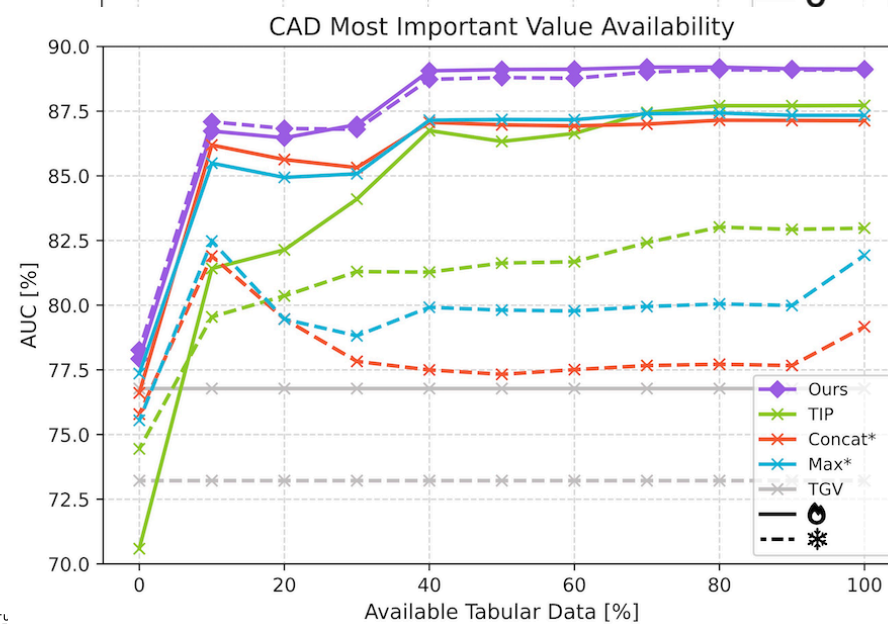
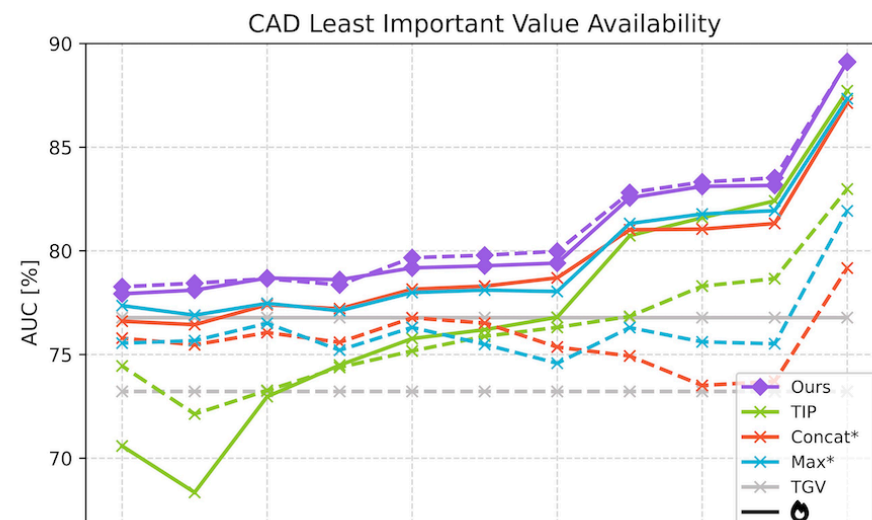
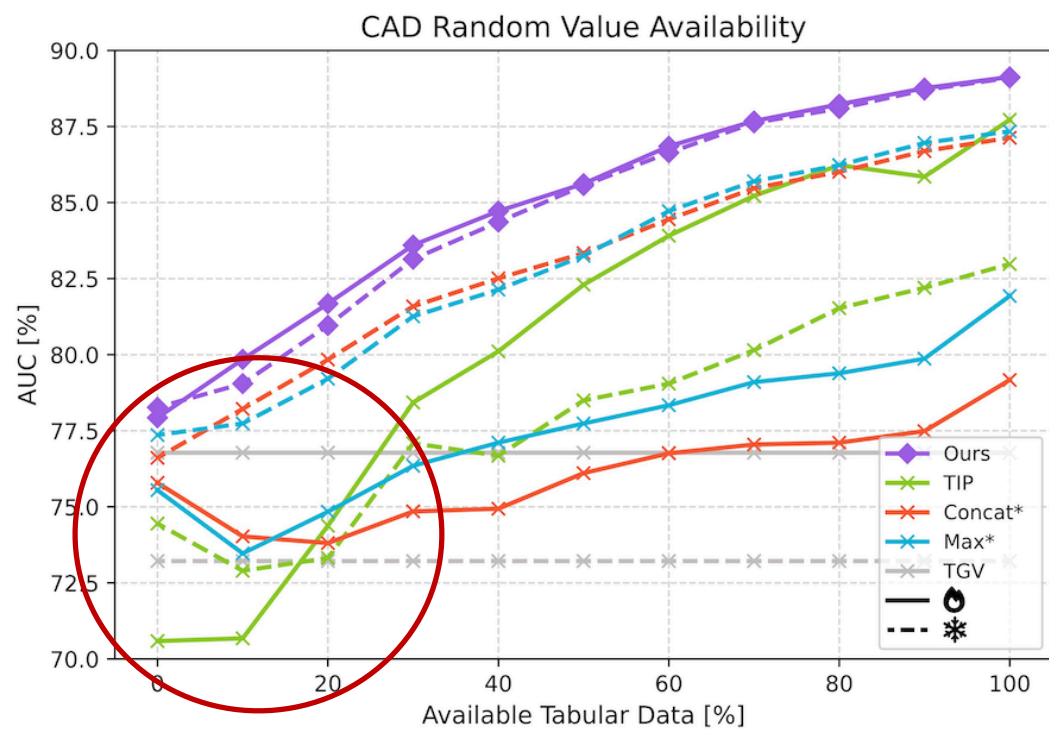


Results

Average performance across the full spectrum of tabular availability (0% - 100%) for Coronary artery diseases (CAD, AUC reported) and Body Mass Index (BMI, MAE reported)

UK Biobank				
	CAD $\uparrow$		BMI $\downarrow$	
	*	🔥	*	🔥
<i>1) End-to-End Methods</i>				
Interact Fuse (Duanmu et al., 2020)	-	61.46	-	2.88
DAFT (Wolf et al., 2022)	-	68.59	-	2.41
<i>2) Fusion Methods</i>				
Max Fuse (Vale-Silva & Rohr, 2021)	74.45	81.38	2.91	1.60
Max Fuse* (Vale-Silva & Rohr, 2021)	77.61	82.90	1.91	1.59
Concat Fuse (Spasov et al., 2019)	75.13	76.33	3.05	1.77
Concat Fuse* (Spasov et al., 2019)	76.10	82.90	2.01	1.79
<i>3) Vision-Tabular Pretraining-based Methods</i>				
<i>Image Only Inference</i>				
MMCL (Hager et al., 2023)	73.52	75.43	2.24	2.22
TGV (Hasny et al., 2025)	73.22	76.78	2.52	2.28
<i>Image-Tabular Inference</i>				
TIP (Du et al., 2024)	78.08	80.49	2.16	2.13
RoVTL (Ours)	84.68	84.91	1.51	1.69

Full Spectrum Robustness



Corruption-Based Augmentation Reduce Performance

	<i>Image-Tabular</i>		<i>Image Only</i>	
	CAD $\uparrow$		LVEF $\downarrow$	
	✱	💧	✱	💧
w/o Augmentations	83.49	83.44	3.13	3.01
Corrupted (MMCL (Hager et al., 2023))	81.07	83.59	3.26	2.99
Gaussian Pert.	71.60	78.64	4.45	3.14
Missing (Ours)	84.68	84.91	3.09	2.99

Conclusion

- **Deployable in Messy Real-World Workflows:** Works across incomplete patient records. Prevents model breakdown caused by missing lab values.
- **Delivers reliable predictions** from raw medical images even when zero tabular clinical data is available.

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Towards real clinical application

Risk Estimation of
Cardiovascular Diseases

Under review

Cardiovascular Disease (CVD) Risk Estimation



CVD is the leading cause of death worldwide (WHO)

→ urgent need for accurate risk prediction



Traditional risk scores [1,2]:

- Rely on **fixed** clinical variables
- Unable to leverage additional data sources



Existing deep learning-based approaches:

- Incorporate more data but are not robust to missing attributes or modalities [3]
- Based on LLMs [4] → computationally expensive



Cine CMR is widely available but underused for CVD risk assessment



Can we employ vision-tabular learning for flexible CVD risk assessment?

[1] 'Agostino Sr, R.B., Vasan, R.S., Pencina, M.J., Wolf, P.A., Cobain, M., Massaro, J.M., Kannel, W.B.: General cardiovascular risk profile for use in primary care: the Framingham heart study. Circulation 117(6), 743–753 (2008)

[2] Collaboration, E.C.R., Group, S.W., et al.: SCORE2 risk prediction algorithms: new models to estimate 10-year risk of cardiovascular disease in Europe. European Heart Journal 42(25), 2439–2454 (2021)

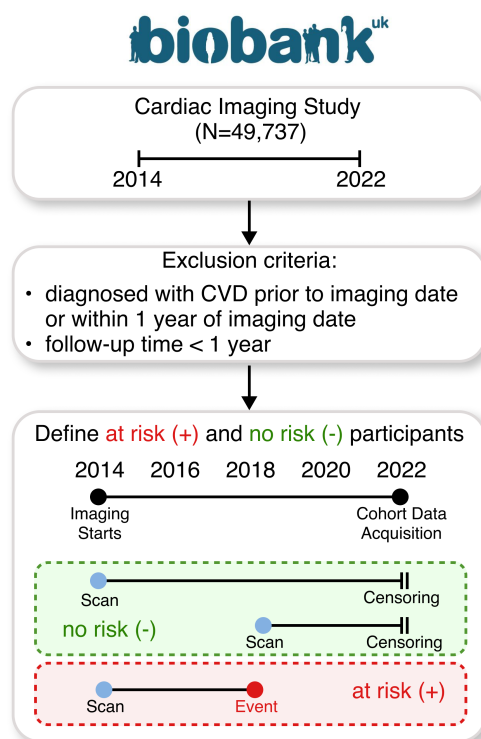
[3] Su, H., Rui, S., Xiang, J., Wu, L., Wang, X.: CTSL: Codebook-based temporal-spatial learning for accurate non-contrast cardiac risk prediction using cine mris. In: International Conference on Medical Image Computing and Computer-Assisted Intervention. pp. 147–157. Springer (2025)

[4] Lübeck, F., Wildberger, J., Träuble, F., Mordig, M., Gatidis, S., Krause, A., Schölkopf, B.: Adaptable cardiovascular disease risk prediction from heterogeneous data using large language models. arXiv preprint arXiv:2505.24655 (2025)

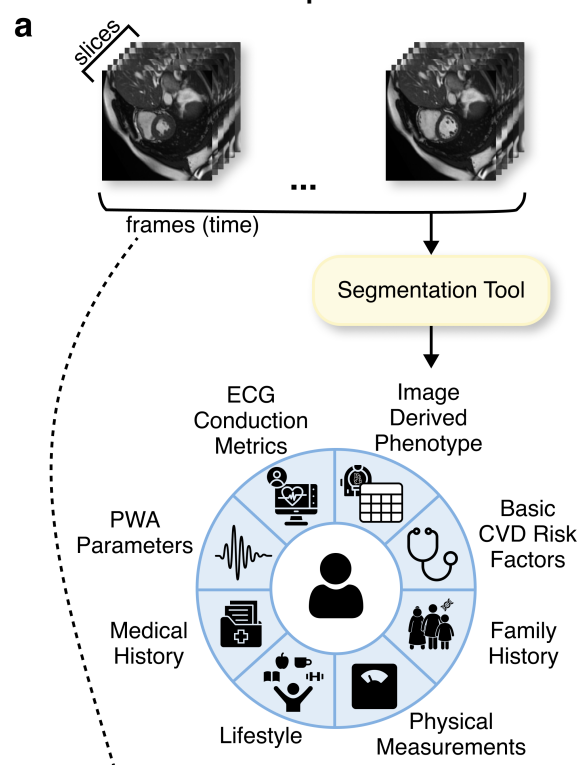
Cardiovascular Disease (CVD) Risk Estimation

Under review

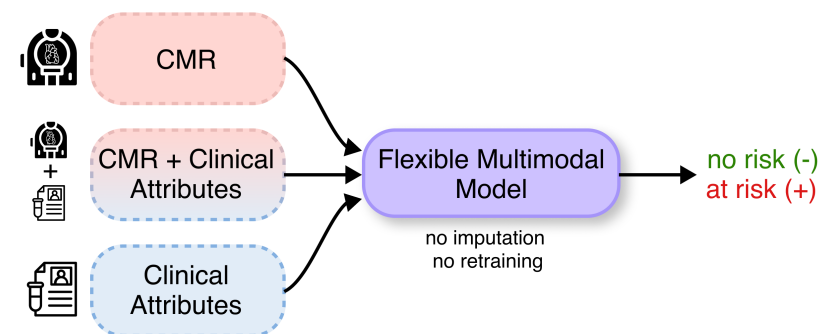
Task



Inputs



Method



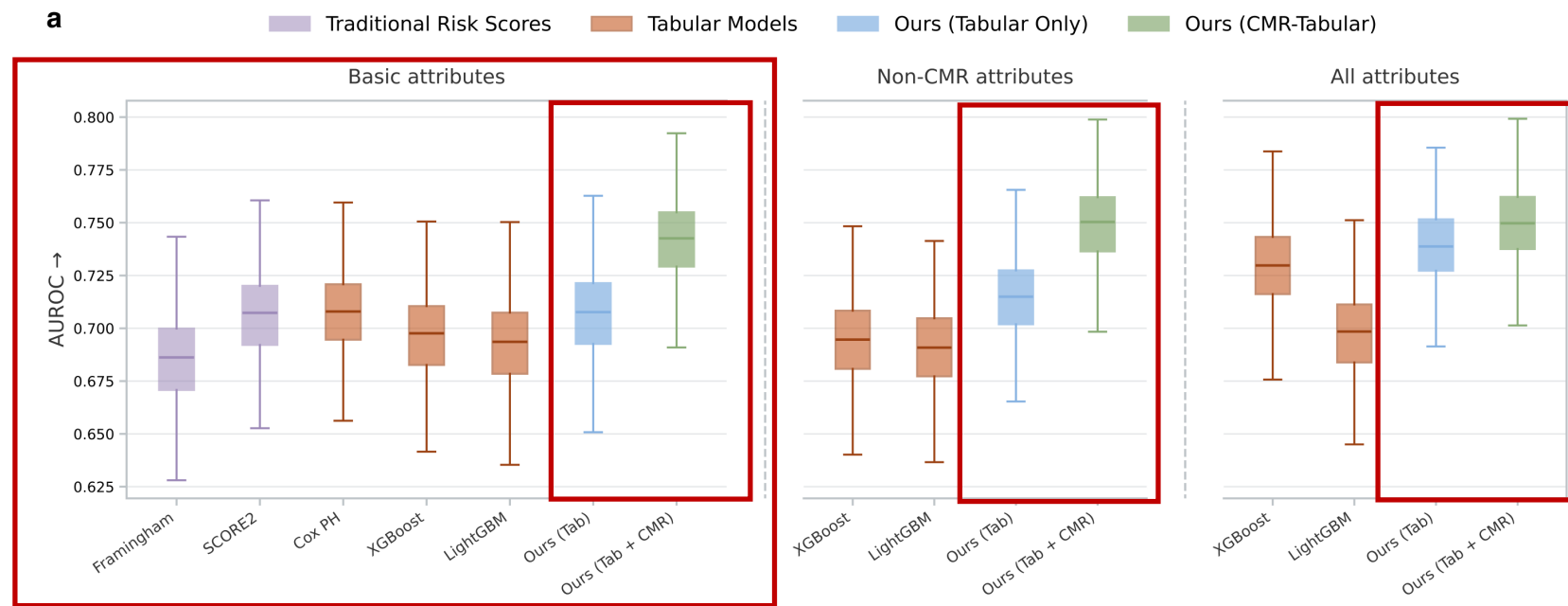
Changes compare to RoVTL:

- Adapt to the case of tabular data only

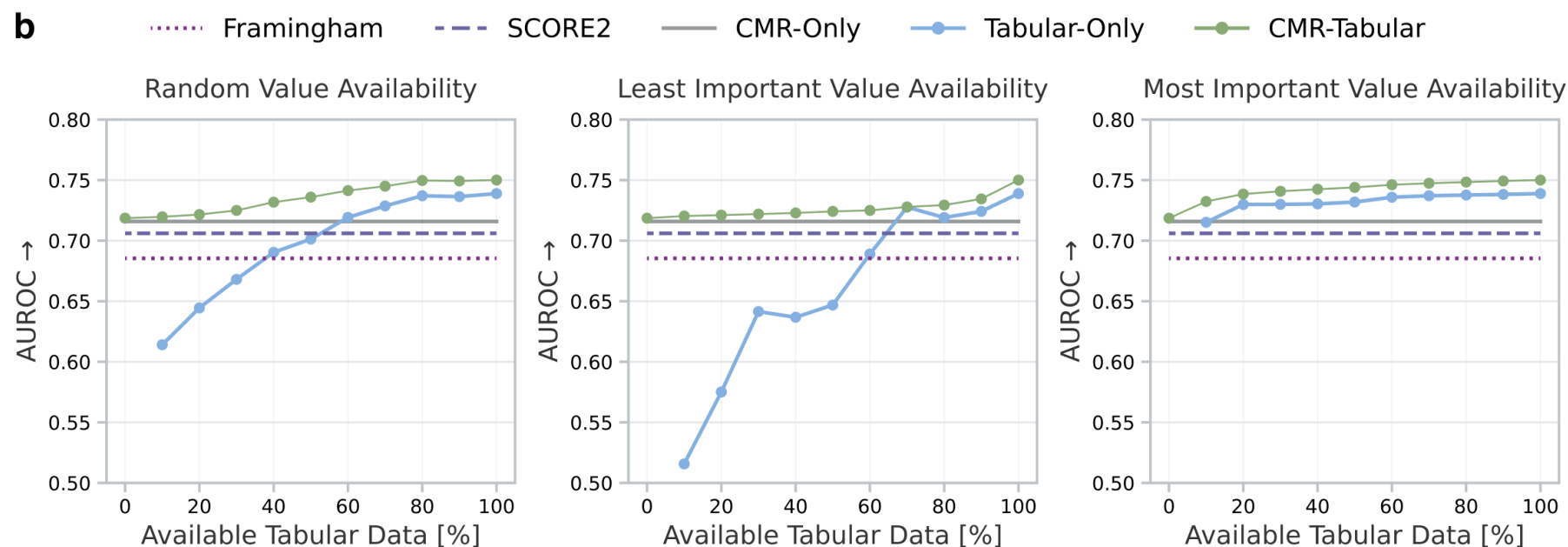
Risk Estimation of CVD

Cine CMR provides strong predictive capability for CVD risk

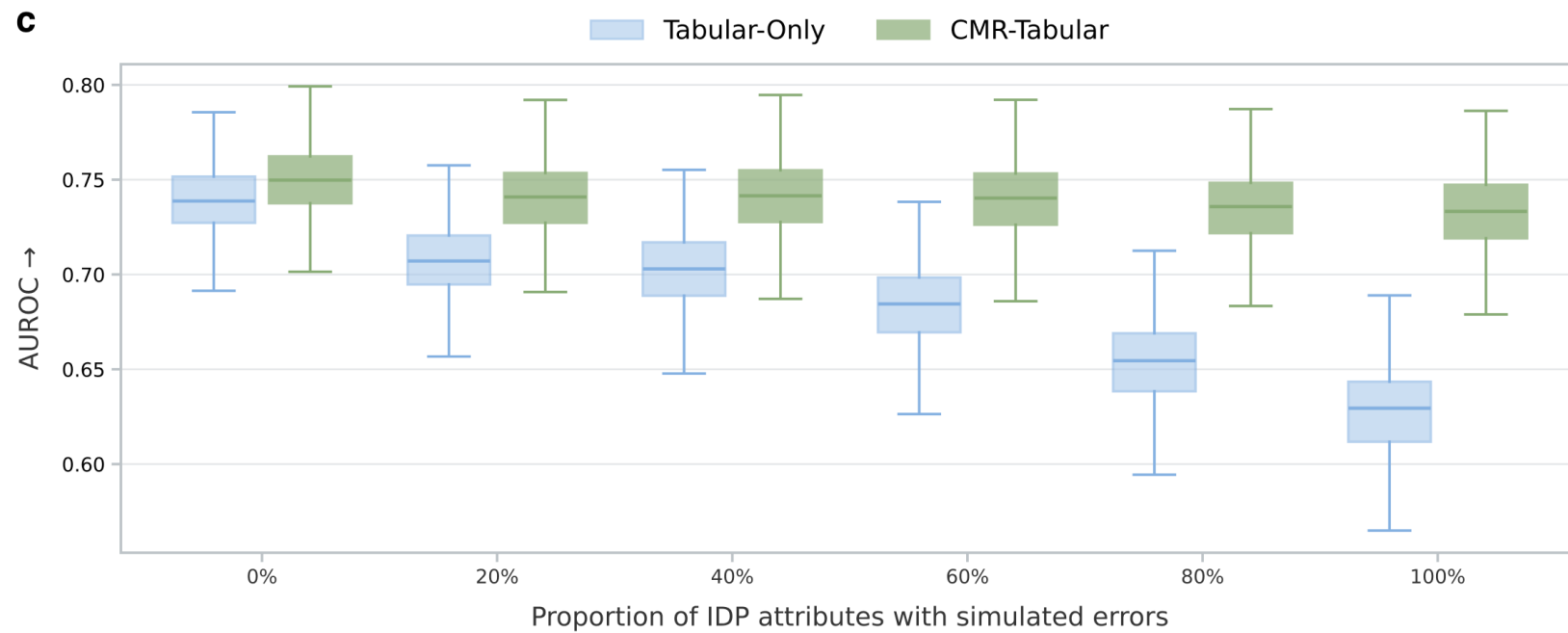
- Establish risk scores use basic attributes set: *age, sex, total cholesterol, high-density lipoprotein (HDL) cholesterol, blood pressure (BP) medication use, systolic BP, smoking status, and diabetes*



CMR increases robustness of the risk estimation



CMR increases robustness of the risk estimation



Conclusion

Takeaway:

Opportunistic cine CMR enhances risk stratification and solves clinical data missingness.

Beyond IDPs: Replaces manual/automated segmentation pipelines (which risk demographic bias and errors).

End-to-End Vision: Processes raw cine CMR directly via a vision encoder with no performance loss.

Efficiency: Eliminates reliance on upstream segmentation without degrading predictive power.

Limitations and Future work:

Cohort Limitations: Retrospective UK Biobank study (predominantly white, volunteer population)
→ Prospective validation in diverse, clinically referred cohorts

Classification Risk: Potential overlap between subclinical/existing disease and long-term risk.

Follow-up Horizon: Shorter follow-up duration compared to standard 10-year clinical risk scores.

Conclusion

Multimodal Reality: Medicine is inherently multimodal

Unifying imaging and tabular clinical data can unlock hidden signals, but needs to handle missing real-world data.

- **Tables Guide Vision (TGV)** [8]
 - **Clinically meaningful training pairs** for visual representation learning
 - **Image only inference**
- **Robust Vision–Tabular Learning (RoVTL)** [9] [10]
 - Bridges **image-only** and **multimodal** inference
 - Outperforms prior vision–tabular methods
 - **Largest gains when tabular data is limited**
 - Potential for clinical applications

➔ Suitable for Biobank-driven learning

Work of Marta Hasny



[8] Hasny, M., Di Folco, M., Bresssem, K., & Schnabel, J. Tables guide vision: Learning to see the heart through tabular data. *In Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*. 2025

[9] Hasny, M., Daza, L., Bresssem, K., Di Folco, M., & Schnabel, J. No Data? No Problem: Robust Vision-Tabular Learning with Missing Values. *International Conference on Machine Learning*. 2026

[10] Hasny, M., Daza, L., Bresssem, K., Di Folco, M., & Schnabel, J. A. (2026). Cine Cardiac MRI Captures Cardiovascular Disease Risk Beyond Established Clinical Risk Factors: Evidence from the UK Biobank. *medRxiv*, 2026-08.

Thank you! Any questions?

Please feel free to reach out for internship or PhD opportunities:

maxime.difolco@telecom-paris.fr

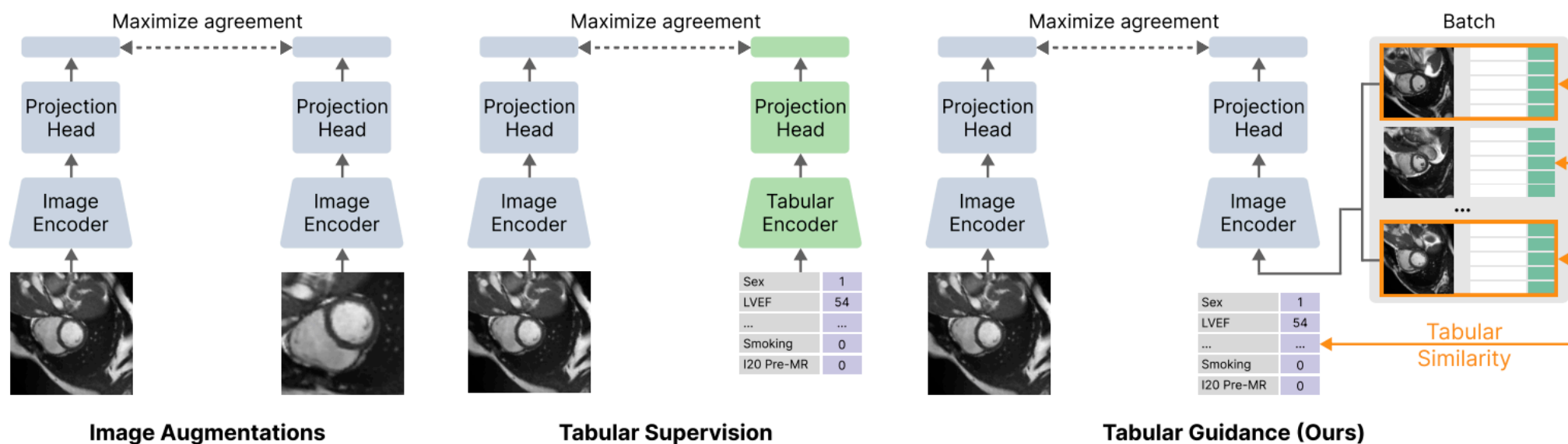


The Effect of Tabular Ranking Loss

a) With Tabular Ranking

b) Without Tabular Ranking

Type of contrastive learning



Presentation overview

Multimodality with Heterogeneous data

Combining Medical Imaging and Clinical Data

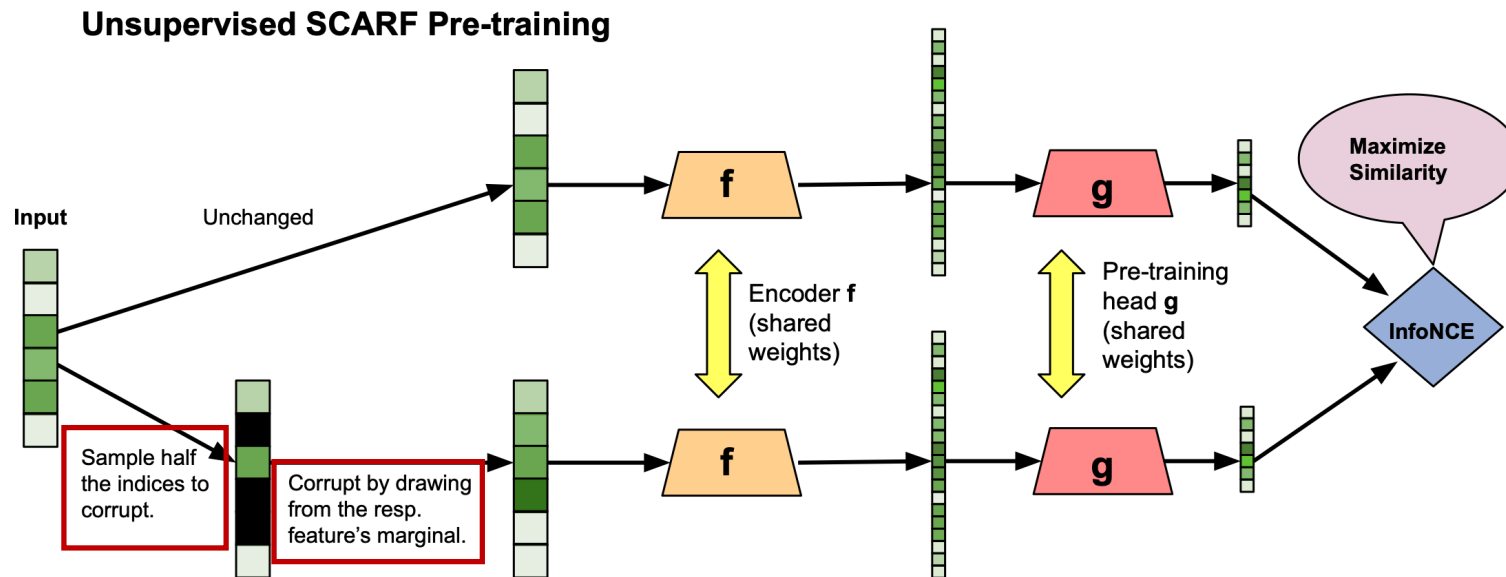
- a. Tables Guide Vision
- b. Robust Vision-Tabular Framework
- c. Risk Estimation of Cardiovascular Diseases

High-Stakes Decision-Making

Towards Trustworthy Decision-Making:
Robustness, Fairness

- a. Mitigate Catastrophic Errors

Augmentation-induced misalignment



1) Unimodal tabular contrastive learning setting [6]

[6] Bahri, D., Jiang, H., Tay, Y., & Metzler, D. Scarf: Self-Supervised Contrastive Learning using Random Feature Corruption. In *International Conference on Learning Representations*.

Presentation overview

Multimodality with Heterogeneous data

Combining Medical Imaging and Clinical Data

- a. Tables Guide Vision
- b. Robust Vision-Tabular Framework
- c. Risk Estimation of Cardiovascular Diseases

High-Stakes Decision-Making

Towards Trustworthy Decision-Making:
Robustness, Fairness

- a. Mitigate Catastrophic Errors

Multipair approcahes

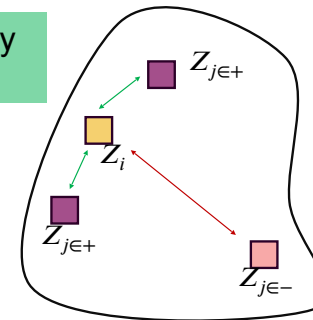
- Set of positive pair P_i is define as:

$$\mathcal{P}_i = \{x_j \mid j \neq i, S_{ij} \geq M_i - h\}$$

h : threshold

M_i : maximum similarity

one-to-many
pairing

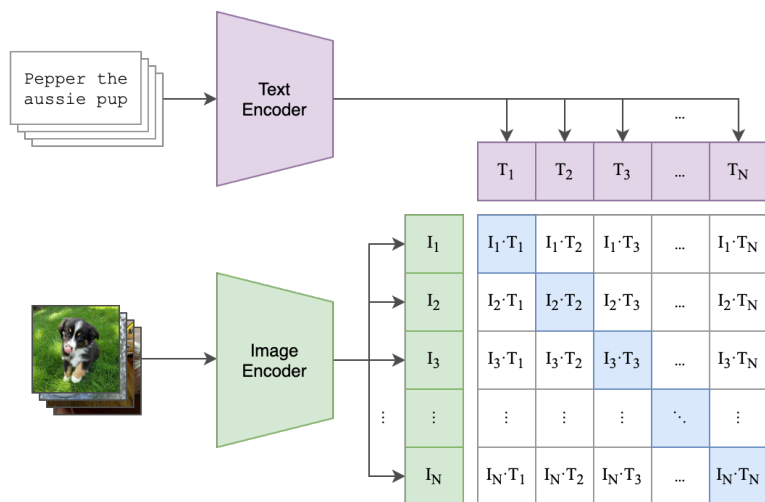


Threshold h		CAD $\uparrow$		LVEF $\downarrow$	
		ZS	FT	ZS	FT
One positive pair only $\rightarrow$	0.0	56.35	72.06	4.90	3.89
	0.05	68.70	76.1	4.08	3.18
	0.1	67.17	74.41	4.37	3.47
Many pairs $\rightarrow$	0.2	67.60	74.63	4.38	3.33

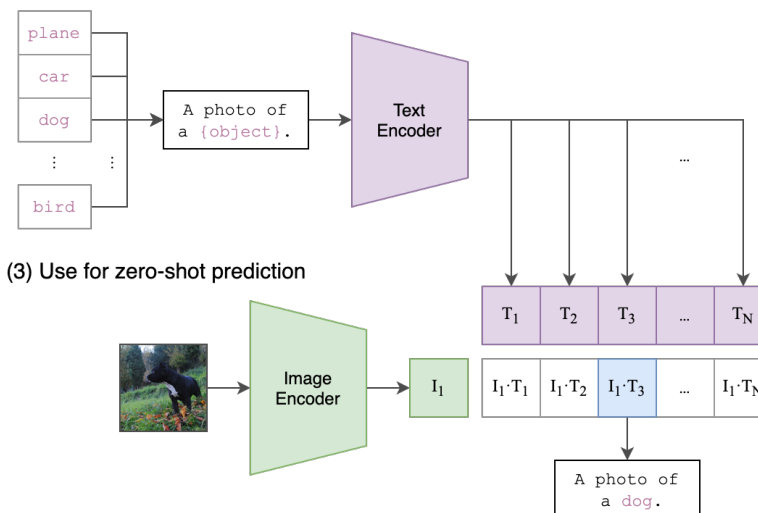
Mitigate Catastrophic Errors

Vision-Language Models (VLMs)

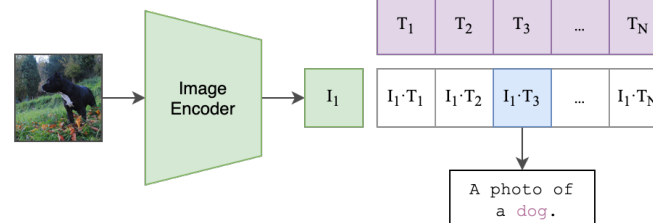
(1) Contrastive pre-training



(2) Create dataset classifier from label text



(3) Use for zero-shot prediction

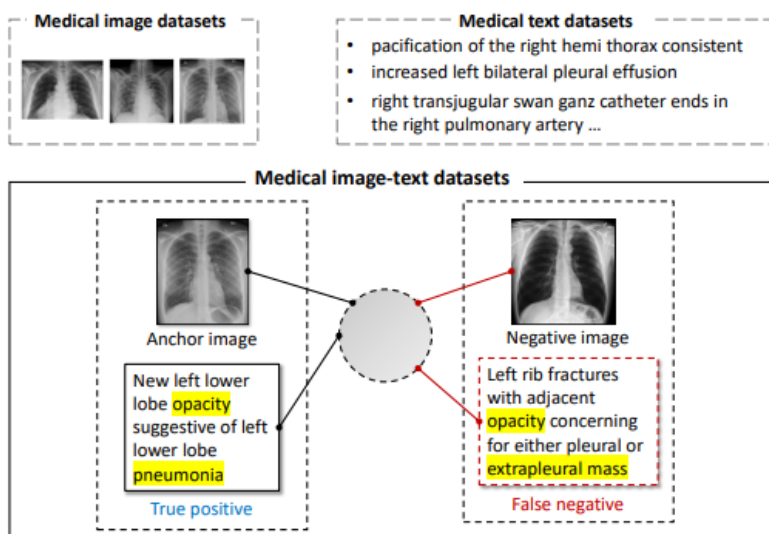


Chen et al., PMLR (2020)

Mitigate Catastrophic Errors

Medical Vision-Language Models

MedCLIP (Wang et al., *EMLNP* 2023)



ACC(STD)	CheXpert-5x200	MIMIC-5x200	COVID	RSNA
CLIP	0.2016(0.01)	0.1918(0.01)	0.5069(0.03)	0.4989(0.01)
CLIP _{ENS}	0.2036(0.01)	0.2254(0.01)	0.5090(<0.01)	0.5055(0.01)
MedCLIP-ResNet	0.5476(0.01)	0.5022(0.02)	0.8472(<0.01)	0.7418(<0.01)
MedCLIP-ResNet _{ENS}	0.5712(<0.01)	0.5430(<0.01)	0.8369(<0.01)	0.7584(<0.01)
MedCLIP-ViT	0.5942(<0.01)	0.5006(<0.01)	0.8013(<0.01)	0.7447(0.01)
MedCLIP-ViT _{ENS}	0.5942(<0.01)	0.5024(<0.01)	0.7943(<0.01)	0.7682(<0.01)

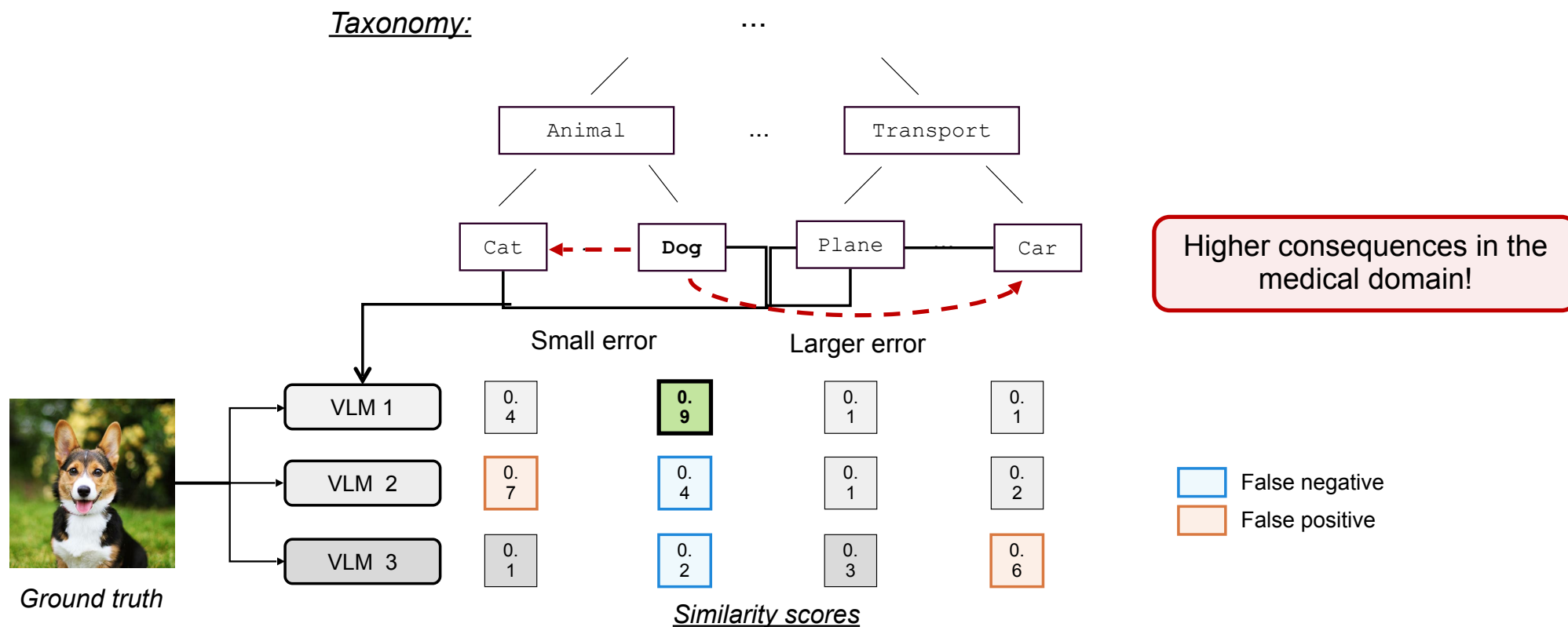
Demonstrate great zero-shot capabilities

Some other models: BioMedCLIP, MedSigLIP, PubMedCLIP ..

! Treat all errors the same way

Mitigate Catastrophic Errors

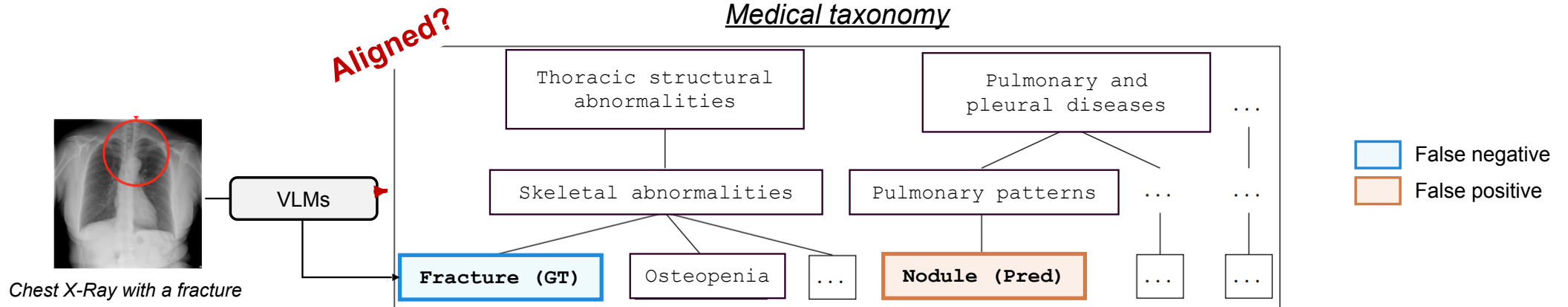
Are errors all the same?



Mitigate Catastrophic Errors

Research question

Are Vision-Language Models align with Medical Taxonomies?



How to quantify the alignment?

Definition of the taxonomy

PadChest-GR Dataset

- Chest X-ray collection containing 4,555 studies
- Four-level medical taxonomy
- **Multi labels classification task**

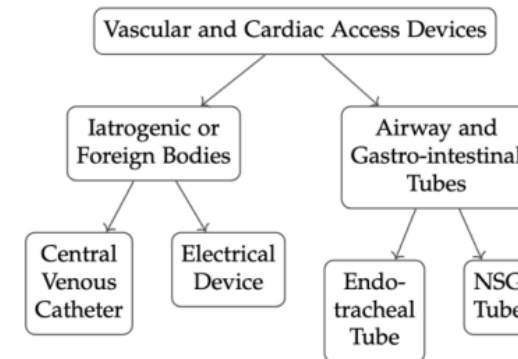
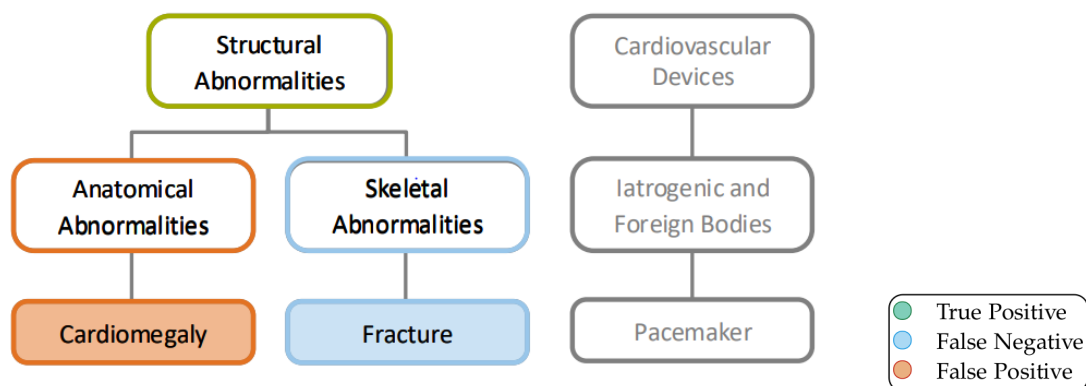


Figure 2 (a) Subtree of the taxonomy, up until the PadChest-GR label groups

Electrical device	pacemaker dual chamber device single chamber device DAI (implantable cardioverter-defibrillator) electrical device
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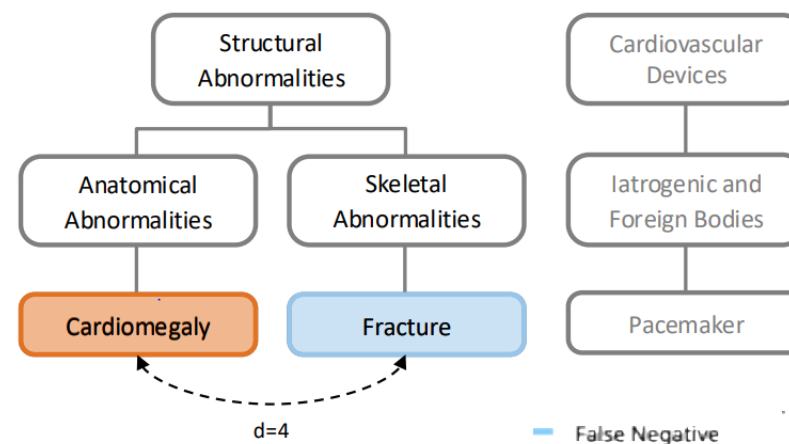
Hierarchy aware metrics

Hierarchical Overlap Score (HOS)



Measures the overlap between the ancestors of the true and predicted labels. (set-based metric)

Hierarchical Distance Score (HDS)

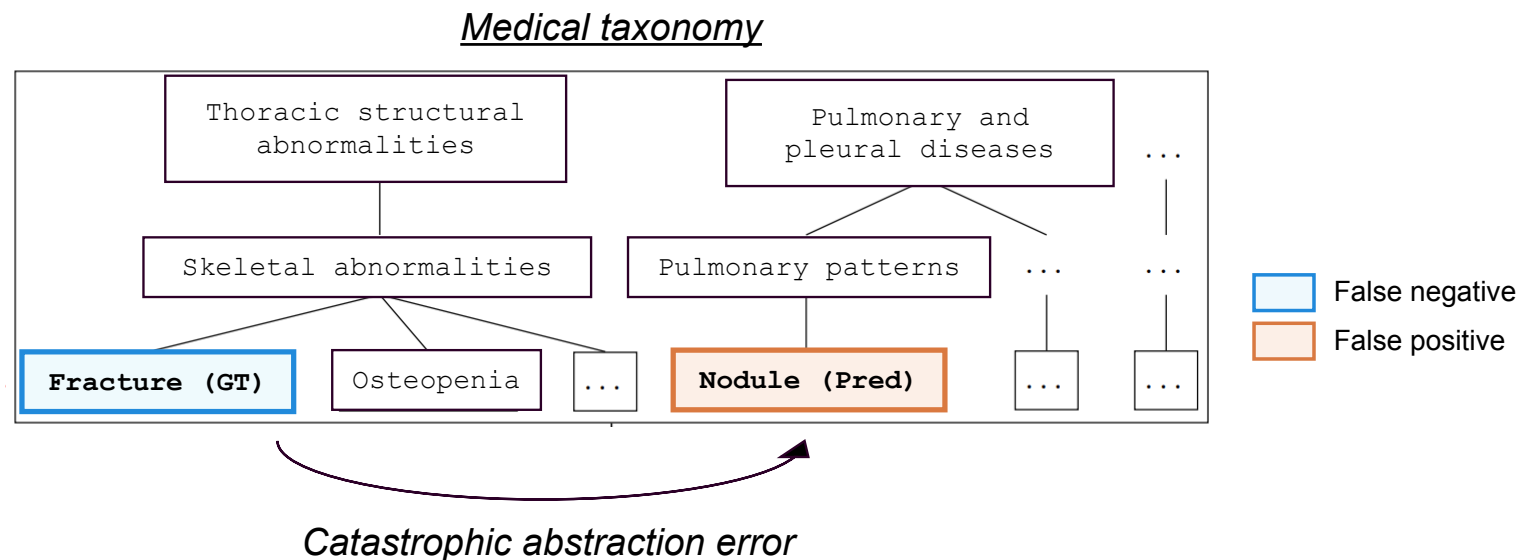


1. **Calculate Score (W):** Score $W = \alpha^d$
2. **Apply Penalty to Both Perspectives**
 - **Recall Perspective (The "Miss"):**
The FN count for Cardiomegaly increases partially
 - **Precision Perspective (The "Mistake"):**
The FP count for Fracture also increases partially

Catastrophic Abstraction Errors

Proposed definition:

"A Catastrophic Abstraction Error (CAE) occurs when predicted and true label sets lie on disjoint branches of the taxonomy, share no common ancestors, and therefore receive zero credit under all hierarchy-aware metrics"



Flat metrics are misleading

Model	$F_1 \uparrow$	HDS $\uparrow$	HOS $\uparrow$	CAE $\downarrow$
<i>Performance-based threshold (Risk $\times$)</i>				
MedSigLIP	22.35 (20.23, 24.26)	21.49 (18.92, 23.90)	43.23 (41.39, 45.02)	19.5
BiomedCLIP	15.36 (13.23, 17.52)	8.71 (6.04, 11.43)	40.30 (38.53, 42.05)	11.6
CLIP Base	11.51 (10.24, 12.83)	3.43 (1.84, 5.35)	33.27 (32.18, 34.49)	0.5
PubMedCLIP	9.68 (8.49, 10.82)	2.48 (0.56, 4.27)	37.64 (36.28, 38.92)	1.5
MedCLIP	14.04 (13.41, 14.69)	0.03 (0.00, 0.07)	24.52 (23.68, 25.45)	0.2

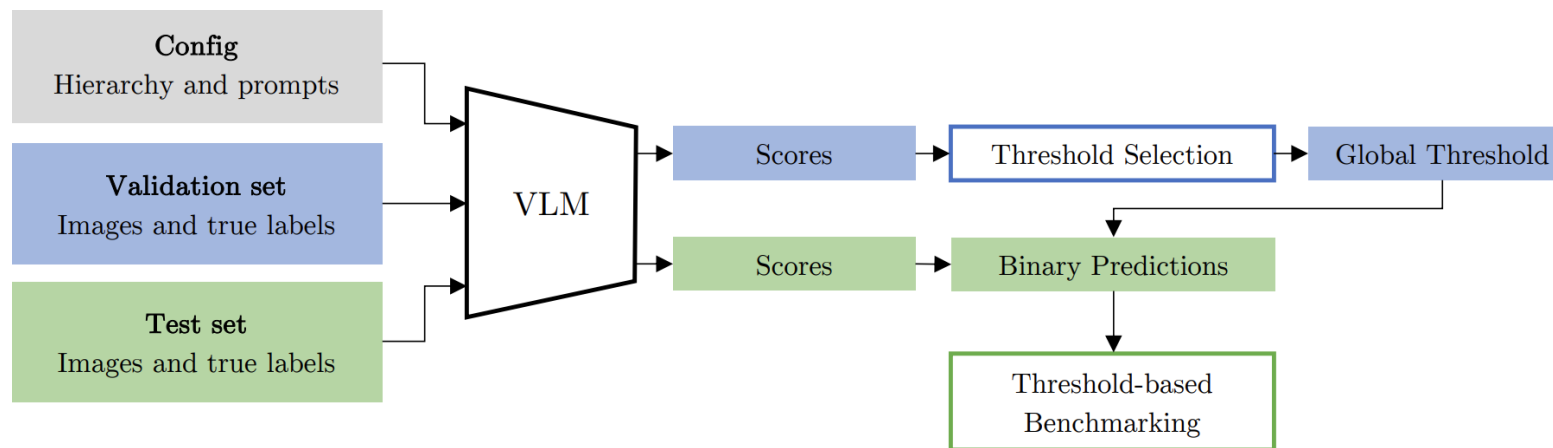
How to mitigate the high CAE rate?

HDS: Hierarchical Distance Score; HOS: Hierarchical Overlap Score; CAE: Catastrophic Abstraction Errors

Mitigate Catastrophic Abstraction Errors

Decision making of VLMs

1. Tune Threshold:
 - Use the validation set to find the optimal global threshold.
2. Evaluate Model:
 - Apply the threshold to the test set for final benchmarking.



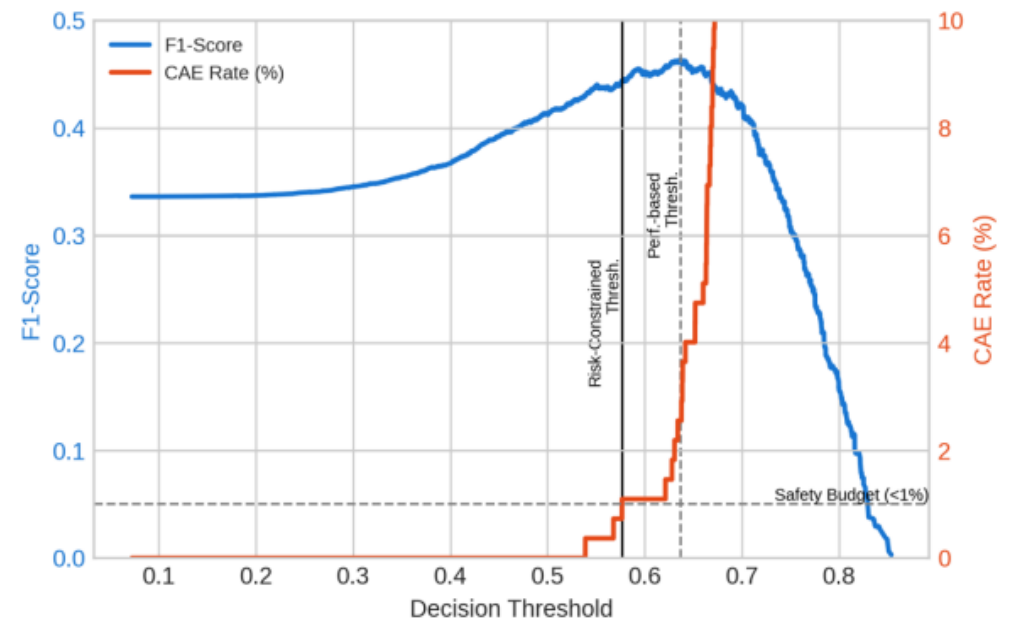
Improving Safety: Thresholding Strategies

1. Performance-based:

Aims to find the single best point for the F1-score.

2. Risk-Constrained (Ours):

First defines a "safe zone" for the CAE rate, then finds the best F1- score within that zone.



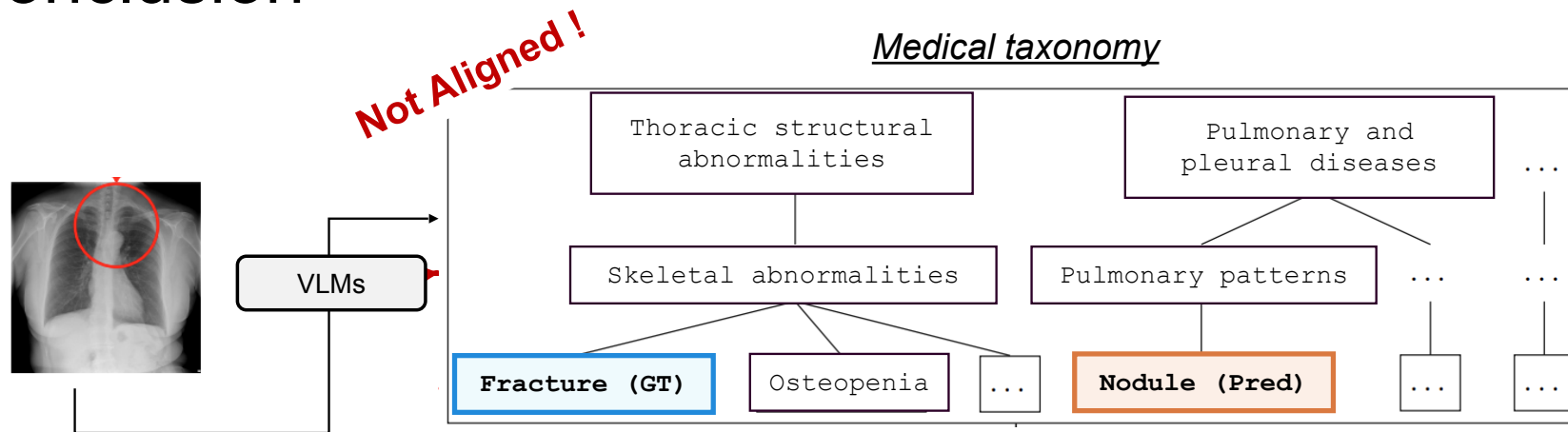
Risk-constrained threshold decreases CAE

Model	$F_1 \uparrow$	HDS $\uparrow$	HOS $\uparrow$	CAE $\downarrow$
<i>Performance-based threshold (Risk $\times$)</i>				
MedSigLIP	22.35 (20.23, 24.26)	21.49 (18.92, 23.90)	43.23 (41.39, 45.02)	19.5
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MedCLIP	14.04 (13.41, 14.69)	0.03 (0.00, 0.07)	24.52 (23.68, 25.45)	0.2
<i>Risk-constrained threshold (Risk $\checkmark$)</i>				
MedSigLIP	17.93 (16.69, 19.19)	8.24 (7.21, 9.58)	31.65 (30.55, 32.78)	0.9
BiomedCLIP	14.75 (13.35, 16.18)	6.98 (5.07, 9.17)	37.57 (36.09, 39.05)	2.4
CLIP Base	10.14 (9.13, 11.16)	2.32 (1.05, 3.45)	34.56 (33.34, 35.76)	1.5
PubMedCLIP	10.09 (8.78, 11.29)	3.61 (1.71, 5.37)	38.55 (37.19, 39.95)	2.2
MedCLIP	13.99 (13.33, 14.65)	0.03 (−0.03, 0.10)	24.45 (23.62, 25.36)	0.7
CLIP Base + TAF (<i>Ours</i>)	21.17 (19.74, 22.51)	15.01 (13.53, 16.80)	33.69 (32.56, 34.81)	1.6

TAF: Taxonomy Aware fine-tuning strategy

HDS: Hierachical Distance Score; HOS: Hierarchical Overlap Score; CAE: Catastrophic Abstraction Errors

Conclusion

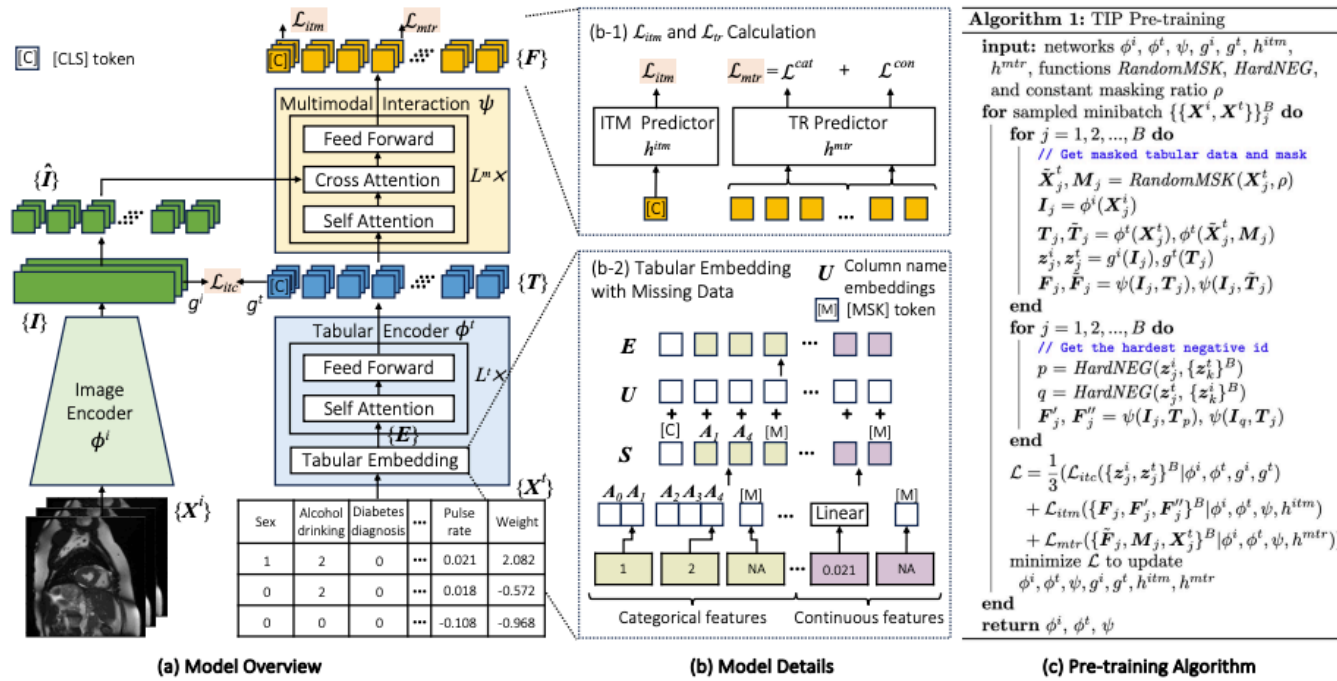


- **VLMs** are not necessarily aligned with the medical taxonomies
- **Classification metrics** are misleading
- **Safety** is a practical choice
 - Can choose to mitigate the catastrophic errors



Work of Ben Schaper

TIP: Tabular Image Pre-training



Du, Siyi, et al. "TIP: Tabular-image pre-training for multimodal classification with incomplete data." ECCV, 2024.

Type of fusion

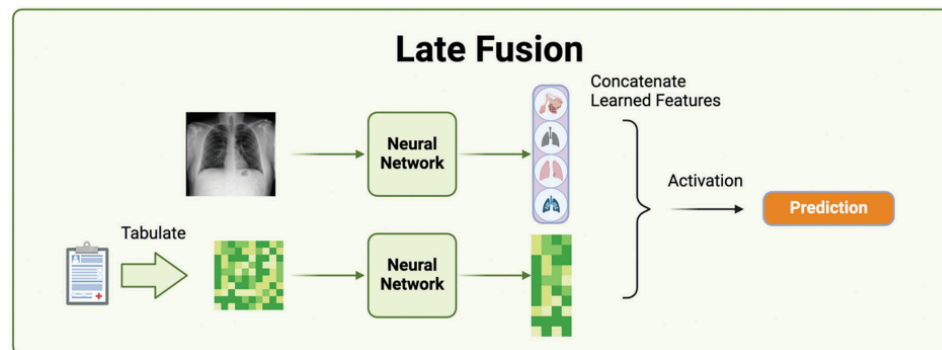
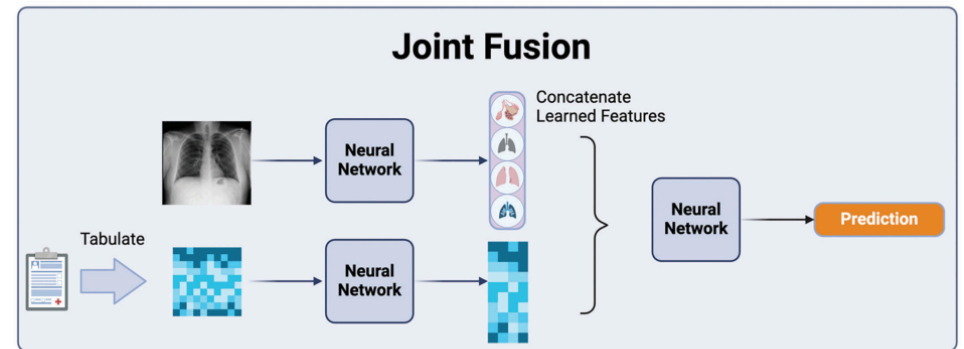
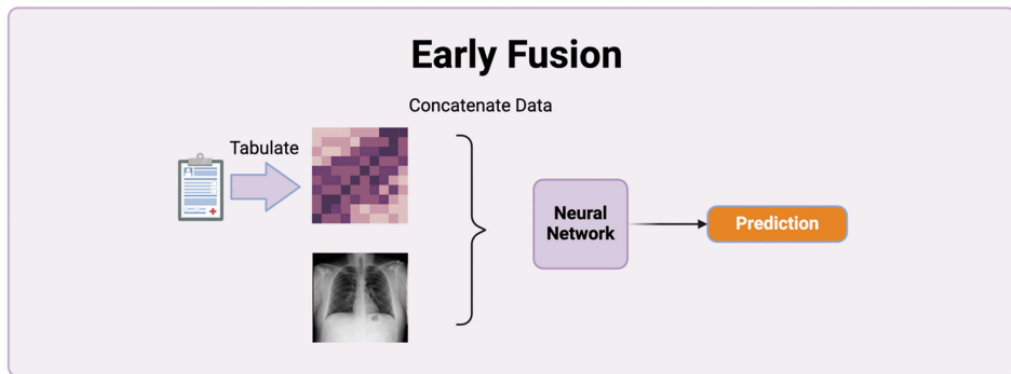


Illustration from: Simon et al. (2025).
Diagnostic and interventional radiology

Tabular foundation models

- Learn rich, transferable feature representations
- **Handle variable input sizes** and heterogeneous features
- Leverage broad, generalizable knowledge from large-scale pretraining

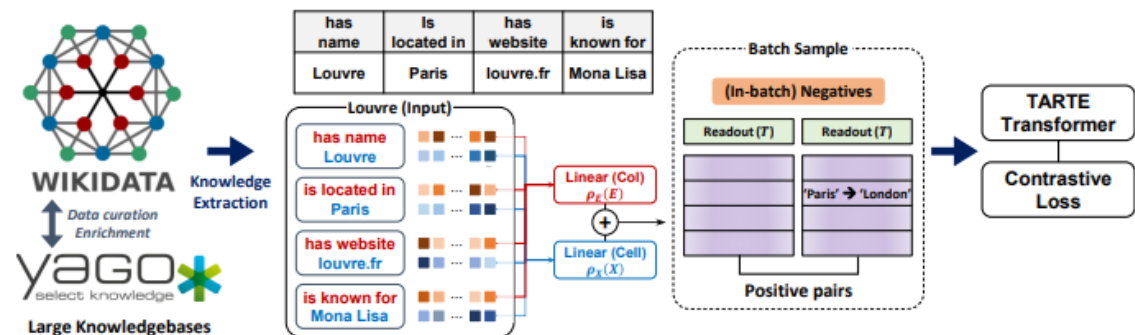


Figure 2: **TARTE Pre-training.** We extract facts from knowledge bases and replicate the input structure in Figure 1. Then, batches are constructed with positive samples by replacing parts of information (e.g., Paris to London). TARTE is trained with contrastive learning using in-batch negatives (Chen et al., 2020b).

Specificity of Tabular Data

- **Heterogeneous variables**

- **Continuous** values (age, lab measurements)
- **Categorical** values (sex, diagnosis)

- **No natural ordering**

- Columns have semantic meaning, not position
- Reordering columns should not change predictions

- **LLMs are not well adapted**

- Assume sequential structure
- Naïve table-to-text representations are fragile

Structured clinical data					
	Sex	Age	Smoker	...	Systolic blood pressure (mmHg)
P1	M	45	No	...	110
P2	W	72	Yes	...	150
...	...	...	...	...	...

Context

Specificity of Tabular Data

- **Heterogeneous variables**

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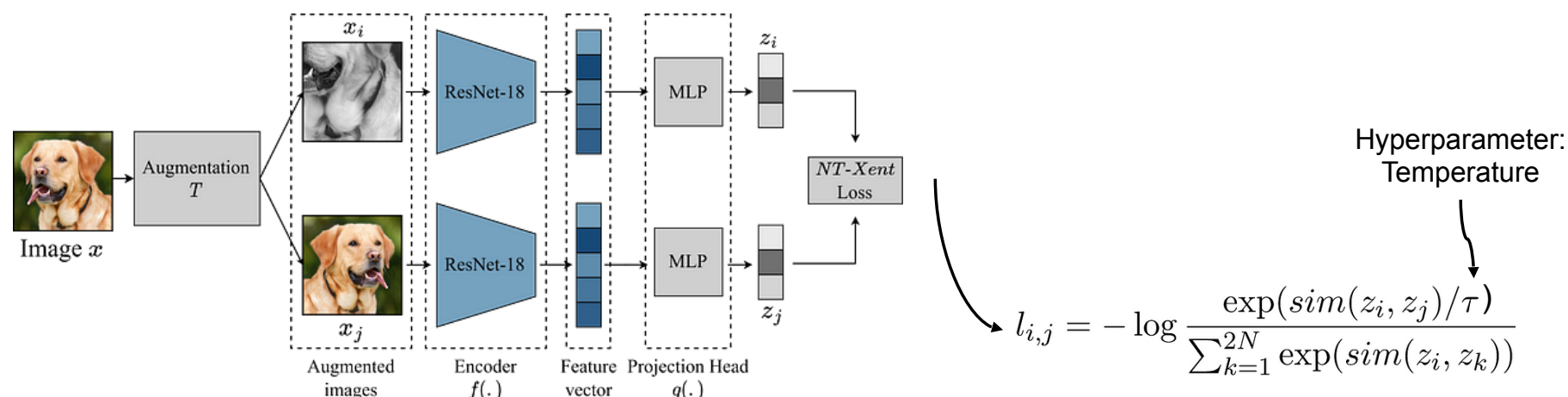
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	Sex	Age	Smoker	...	Systolic blood pressure (mmHg)
P1	M	45	No	...	110
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...	...	...	...	...	...

A simple framework for contrastive learning: SimCLR_[1]

- SimCLR



SimCLR framework. Image by author.

NT-Xent Loss encourages:

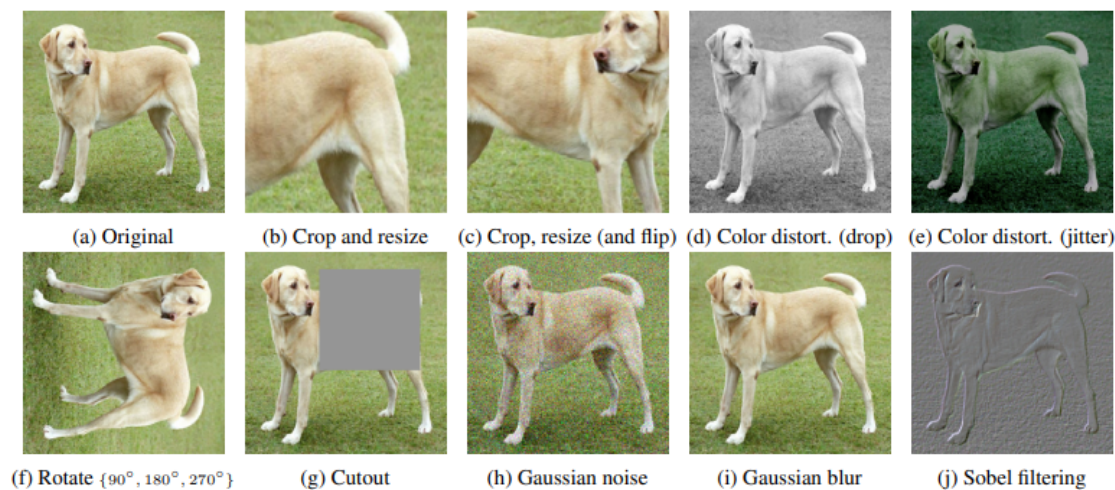
- similar representations for the same image
- dissimilar representations for different images

[1] - Chen, Ting, et al. "A simple framework for contrastive learning of visual representations." *International conference on machine learning*. Pmlr, (2020).

Combining Medical Imaging and Clinical Data

Data augmentations

- SimCLR



Data augmentation transformations. Image by author.

[1] - Chen, Ting, et al. "A simple framework for contrastive learning of visual representations." *International conference on machine learning*. PmLR, (2020).

Methods

AMORE: Adaptive Multimodal Outcome Risk Estimator

→ AMORE supports CMR-only, tabular-only, and CMR-tabular predictions, including tabular data missingness scenarios.



AMORE: Adaptive Multimodal Outcome Risk Estimator



Tabular encoder: TabPFN [1]

- Operates on in-context learning
- Must be conditioned on a task-specific training set to obtain meaningful embeddings
- Result: embedding and tabular-only predictions obtained together

[1] Hollmann, Noah, et al. "Tabpfn: A transformer that solves small tabular classification problems in one shot." *arXiv preprint arXiv:2207.01848* (2022).

Methods

AMORE: Adaptive Multimodal Outcome Risk Estimator

→ AMORE supports CMR-only, tabular-only, and CMR-tabular predictions, including tabular data missingness scenarios.

